Sound: pitch and amplitude

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a drawing or explanation of a longitudinal wave to distinguish pitch from sound level.

SUCCESS CRITERIA - I can...

1. Describe compressions and rarefactions.
2. Connect pitch primarily with frequency.
3. Separate particle motion from sound propagation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A wave enters a medium where its speed is **3** times as large, at a small angle below the critical angle. Does the transmitted ray bend toward or away from the normal?

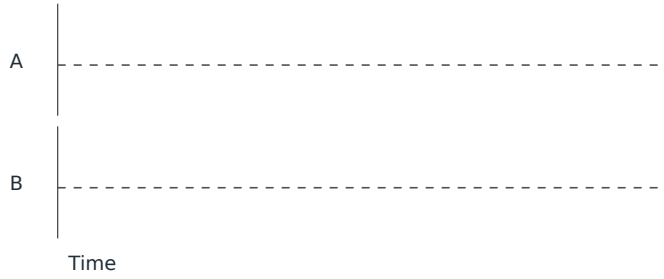
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $T = \mathbf{1}/f$; $v = f\lambda$.
Higher f generally means higher pitch. Amplitude affects intensity with the other parts of the setup kept the same, not frequency.

Worked reasoning: A tuning fork makes **440 cycles** each second, so its frequency is **440 Hz**. A louder version can have the same frequency but greater amplitude.

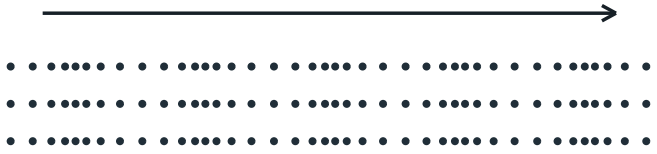
Your turn: Draw two displacement-time traces with equal periods and different amplitudes.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark compressions and rarefactions in the particle picture. Do air particles travel from speaker to listener with the disturbance?

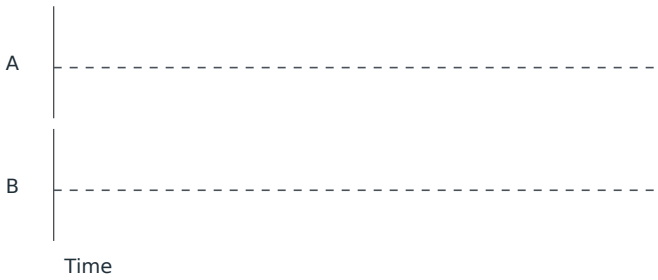


Label dense and sparse regions.

Marzano check - circle: ① ② ③

04 Apply the physics

A tone generator plays A at **200 Hz** and B at **400 Hz**. Which has higher pitch?
1. **Find how many cycles each** produces in **2 s**.
2. **Sketch time traces** with equal amplitudes to **show the frequency difference**.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Turning up a speaker’s volume does not normally double a song’s pitch. **Explain which wave property mainly** changes.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A refrigerator uses **24 J** electrical work while moving heat from its colder interior to a warmer room. **Explain why this** is not spontaneous cold-to-hot heat flow.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.

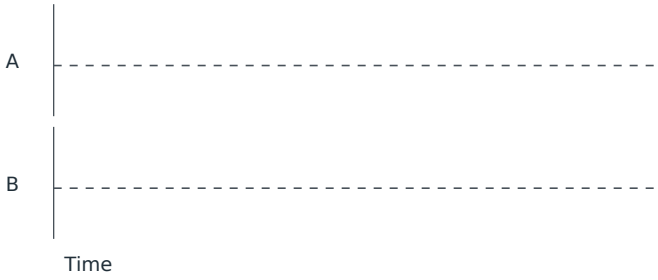
Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

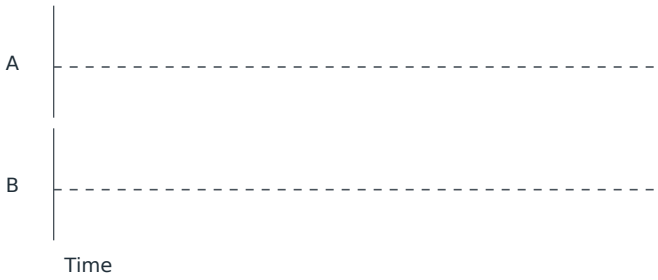
Is sound in air mainly transverse or longitudinal? **State how particles move relative** to propagation.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Design two sound changes: one changes pitch only and one changes amplitude only.
2. **Describe how their graphs differ**.



Marzano check - circle: 1 2 3